

Mastering Quantum Mechanics — Problem 9.8

Ground state and first excited state of a potential

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Problem 9.8

Ground state and first excited state of a potential.

Consider an even potential of height V_0 except for a dip around $x = 0$, where it is accurately quadratic, $V(x) \simeq \frac{1}{2}\alpha x^2$ ($\alpha > 0$).

1. Use the harmonic oscillator to estimate the ground-state energy, and find the inequality among $\alpha, V_0, m, \hbar$ required for the estimate to be accurate.
2. Build an even potential from two such copies with well-separated centers at $\pm x_0$. Give the approximate energies of the ground state and first excited state, sketch the wave functions, and state the inequality among $x_0, \alpha, \hbar, m$ required for accuracy.
3. Combine the two inequalities into the form $\dots \ll \alpha \ll \dots$.

Hints & Orientation

- **Part 1.** Match the dip to an oscillator by comparing $\frac{1}{2}\alpha x^2$ with $\frac{1}{2}m\omega^2 x^2$ (Eq. (9.1.8)) to read off ω ; then use $E_n = \hbar\omega(n + \frac{1}{2})$ (Eq. (9.2.30)) at $n = 0$. The estimate is valid only if the state lives *deep in the dip*, i.e. its energy is far below the plateau: $E_0 \ll V_0$. Turn that into an **upper bound on α** .
- **Part 2.** Don't solve a new equation: each far-apart dip hosts its own localized ground state $\varphi_{L,R}$ (the oscillator ground state shifted to $\pm x_0$). The two lowest states of the even potential are the symmetric and antisymmetric combinations (bonding/antibonding picture); to this accuracy they are degenerate at $\frac{1}{2}\hbar\omega$. Node counting fixes which is the ground state. "Well separated" means the oscillator width $L_0 = \sqrt{\hbar/m\omega}$ (Eq. (9.1.14)) must be $\ll x_0$ — a **lower bound on α** .
- **Part 3.** Stack the upper bound (Part 1) and the lower bound (Part 2).
- **Targets to check:** $\omega = \sqrt{\alpha/m}$; the bounds scale as $\alpha \sim V_0^2$ (high side) and $\alpha \sim x_0^{-4}$ (low side).

Solution

Part 1 — single dip

Near $x = 0$ the potential is $V \simeq \frac{1}{2}\alpha x^2$. Identifying this with the oscillator potential $\frac{1}{2}m\omega^2 x^2$ gives

$$m\omega^2 = \alpha \quad \implies \quad \omega = \sqrt{\frac{\alpha}{m}}. \quad (1)$$

The ground-state energy estimate is therefore

$$E_0 \simeq \frac{1}{2}\hbar\omega = \frac{1}{2}\hbar\sqrt{\frac{\alpha}{m}}. \quad (2)$$

Validity. The harmonic approximation holds only while the particle stays in the region where $V \approx \frac{1}{2}\alpha x^2$, i.e. the ground-state energy must lie well below the plateau height, $E_0 \ll V_0$:

$$\frac{1}{2}\hbar\sqrt{\frac{\alpha}{m}} \ll V_0 \implies \alpha \ll \frac{4mV_0^2}{\hbar^2} \quad \left(\text{order of magnitude } \alpha \ll \frac{mV_0^2}{\hbar^2}\right). \quad (3)$$

Part 2 — two wells at $\pm x_0$

Each isolated dip supports a localized ground state, namely the oscillator ground state shifted to its center,

$$\varphi_L(x) = \varphi_0(x + x_0), \quad \varphi_R(x) = \varphi_0(x - x_0), \quad (4)$$

each with energy $\simeq \frac{1}{2}\hbar\omega$. Because the full potential is even, the two lowest stationary states are the symmetric and antisymmetric combinations

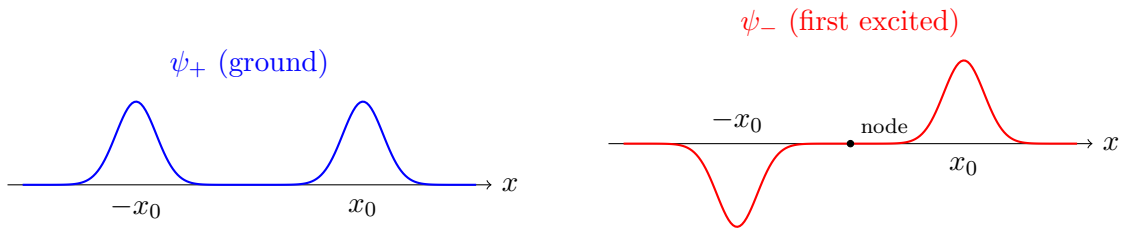
$$\psi_+ = \frac{\varphi_L + \varphi_R}{\sqrt{2}} \quad (\text{even, no node}), \quad \psi_- = \frac{\varphi_L - \varphi_R}{\sqrt{2}} \quad (\text{odd, one node at } x = 0). \quad (5)$$

By node counting the **ground state** is ψ_+ and the **first excited state** is ψ_- . With the wells well separated the overlap is negligible, so both energies coincide with the single-well value to the accuracy of this estimate:

$$E_{\text{gs}} \simeq E_{1\text{st}} \simeq \frac{1}{2}\hbar\sqrt{\frac{\alpha}{m}}. \quad (6)$$

(The true splitting is exponentially small in the well separation — a tunneling effect beyond this approximation.)

Sketches.



Validity. “Well separated” means each localized state, of width $L_0 = \sqrt{\hbar/m\omega}$ (Eq. (9.1.14)), must not overlap its partner: $L_0 \ll x_0$. Since

$$L_0^2 = \frac{\hbar}{m\omega} = \frac{\hbar}{\sqrt{m\alpha}} \implies L_0^4 = \frac{\hbar^2}{m\alpha}, \quad (7)$$

the condition $L_0^4 \ll x_0^4$ becomes

$$\alpha \gg \frac{\hbar^2}{mx_0^4}. \quad (8)$$

Part 3 — combined inequality

The upper bound from Part 1 and the lower bound from Part 2 sandwich α :

$$\boxed{\frac{\hbar^2}{mx_0^4} \ll \alpha \ll \frac{mV_0^2}{\hbar^2}} \quad (9)$$

(keeping only orders of magnitude). Such a window exists provided $\hbar^2/(mx_0^4) \ll mV_0^2/\hbar^2$, i.e. $x_0^4 \gg \hbar^4/(m^2V_0^2)$ — the wells must be far apart compared with the natural quantum length set by V_0 .

Checks

- **Dimensions.** $[\alpha] = \text{energy}/\text{length}^2$; both bounds $\hbar^2/(mx_0^4)$ and $mV_0^2/\hbar^2$ carry exactly those units.
- **Physical scaling.** A deeper plateau (larger V_0) tolerates a stiffer dip (larger α); wider separation (larger x_0) relaxes the lower bound — both as expected.
- **Degeneracy.** The near-degenerate ground/first-excited pair is the hallmark of a symmetric double well; the splitting only appears once tunneling is included, consistent with ignoring overlap here.